

Requirement Analysis: Palindrome Checker

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1. Introduction

1.1 Project Purpose

The "Palindrome Checker" is a software application whose primary purpose is to analyze a string of text provided by a user and determine if that string is a palindrome.

1.2 Project Scope

The system will provide a simple user interface (UI) for a user to input a single string. It will then process this string and display a clear, binary result (e.g., "Yes, it is a palindrome" or "No, it is not a palindrome").

This project is focused *only* on the logic and UI for the check. It will *not* include features such as:

- User accounts or login
- A database to store previous results
- Batch processing of multiple strings

1.3 Definitions

- **Palindrome:** A word, phrase, number, or other sequence of characters that reads the same backward as forward.
- **Alphanumeric Characters:** The letters of the alphabet (A-Z, a-z) and numerical digits (0-9).
- **Non-Alphanumeric Characters:** Any character that is not a letter or a number. This includes spaces, punctuation, and special symbols (e.g., !, ?, ,, :,).
- **UI (User Interface):** The visual elements of the application that the user interacts with (e.g., a text box and a button).

2. Stakeholders

- **Project Owner:** The end-user or requestor who needs the tool.
- **User:** Any individual who wants to check if a string is a palindrome.
- **Developer:** The individual or team responsible for building and testing the application.

3. Functional Requirements (FR)

These are the core "what the system must do" requirements.

FR-1: User Input

- The system shall provide a text input field where the user can type or paste a string.
- The system shall provide a button (e.g., "Check") or similar mechanism for the user to

submit the input string for analysis.

FR-2: String Processing (Normalization) To ensure an accurate and flexible check, the system *must* normalize the input string before analysis.

- **FR-2.1 (Case Insensitivity):** The system shall convert the entire input string to a single, uniform case (e.g., all lowercase).
 - *Example:* "Racecar" becomes "racecar".
- **FR-2.2 (Character Filtering):** The system shall remove all non-alphanumeric characters from the input string.
 - *Example:* "A man, a plan, a canal: Panama" becomes "amanaplanacanalpanama".
 - *Example:* "Eva, can I see bees in a cave?" becomes "evacaniseebeesinacave".

FR-3: Palindrome Logic

- The system shall compare the normalized string (from FR-2) against the reverse of that same normalized string.

FR-4: Output / Result Display

- **FR-4.1 (Positive Result):** If the normalized string is identical to its reverse, the system shall display a clear, unambiguous message indicating the input *is* a palindrome (e.g., "Yes, this is a palindrome!").
- **FR-4.2 (Negative Result):** If the normalized string is *not* identical to its reverse, the system shall display a clear, unambiguous message indicating the input *is not* a palindrome (e.g., "No, this is not a palindrome.>").
- **FR-4.3 (Feedback):** The result message shall be displayed prominently to the user after the "Check" button is pressed.

FR-5: Edge Case Handling

- **FR-5.1 (Empty String):** If the user submits an empty string or a string containing *only* non-alphanumeric characters (which results in an empty normalized string), the system shall consider it a palindrome.

4. Non-Functional Requirements (NFR)

These requirements define the *quality* and *characteristics* of the system.

- **NFR-1 (Usability):** The user interface shall be simple, clean, and intuitive, requiring no instructions for a user to operate.
- **NFR-2 (Performance):** The palindrome check and result display shall be near-instantaneous (e.g., under 1 second) for all reasonable input lengths (e.g., up to 20,000 characters).
- **NFR-3 (Technology):** The application shall be a self-contained, web-based tool running entirely in the client's browser (e.g., built with HTML, CSS, and JavaScript). No server-side processing is required.
- **NFR-4 (Compatibility):** The application must render correctly and be fully functional on the latest versions of modern web browsers (e.g., Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge) on both desktop and mobile devices.

5. Assumptions and Constraints

- **Assumption-1:** The definition of "palindrome" for this project *intentionally* ignores case, punctuation, and spacing to accommodate phrases, not just single words.
- **Constraint-1:** The application will not be connected to a database. It will not save, log, or store any user inputs or results after the user naviges away from the page.

- **Constraint-2:** The primary character set for testing will be the English alphabet (A-Z) and Arabic numerals (0-9). Complex Unicode characters (e.g., emojis, complex diacritics) that may have ambiguous reversal or case-conversion rules are not a primary focus.